2ND SEMESTER LAB REPORT

TITLE : Electrical Circuit Analysis Lab



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Submitted To:

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EXPERIMENT NO 8

Lab Title: Lab – Object Oriented Programming

Objective: Operators Overloading in OOP

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 18/March/2025 Signature:

LABORATORY EXPERIMENT NO. 8

TO VERIFY MESH ANALYSIS FOR DC CIRCUITS BOTH ON HARDWARE AND SOFTWARE

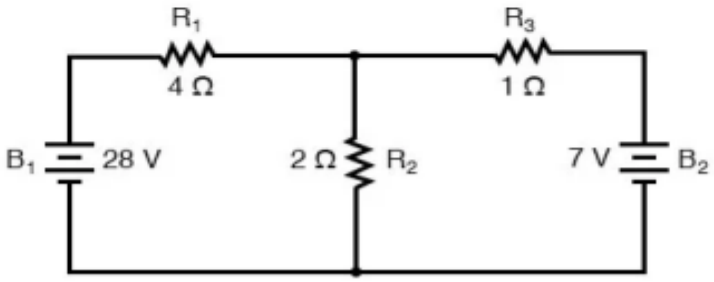
OBJECTIVE:

1. To verify experimentally the calculated currents found by using the mesh current method.
2. To verify the same circuits on Proteus software.

Steps for Mesh Analysis:

1. Assign mesh currents $i_1, i_2, .., i_n$ to the n meshes.
2. Solve for easy meshes.
3. Apply KVL to each of the n meshes.
4. Use Ohm's law to express the voltages in terms of the mesh currents.
5. Solve the resulting n simultaneous equations to get the mesh currents.

Lets solve an example using mesh analysis:



Lab Tasks:

Lab task 1:

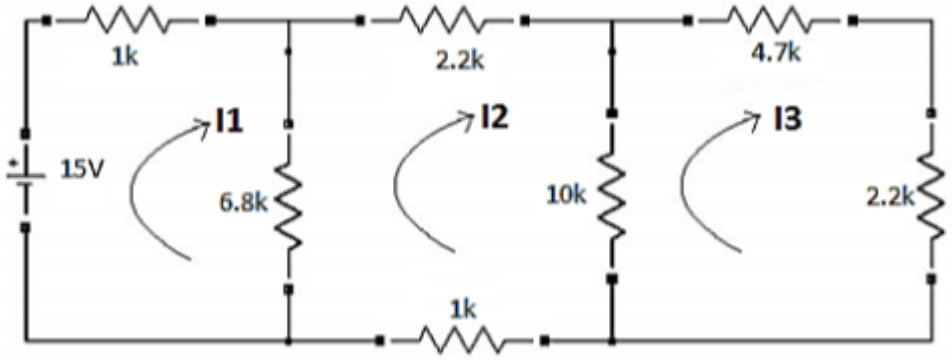
1. Solve using Mesh Analysis:

a) Take seven resistor, according to the values mentioned in the figure.

b) Now construct the circuit as shown in the figure. After verifying that the circuit is properly patched on the breadboard, switch on the trainer.

c) Using mesh analysis, solve for each loop. Record the values and calculations. Measure the mesh currents using DMM. Record the values in data table.

d) Now turn off the power of trainer and DMM.



Mesh Currents	I (Calculated)	I (Measured)
Mesh 1	3.334mA	3.002mA
Mesh 2	1.6mA	1.57mA
Mesh 3	0.9mA	0.88mA

e) Now calculate and measure the current and voltage for each resistor and write down in the following table:

Resistors	V (Calculated)	V (Measured)	I (Calculated)	I (Measured)
1K ohm	3.32V	3.11V	3.02mA	3.32mA
2.2K ohm	11.7V	10.91V	1.32mA	1.6mA
4.7K ohm	3.53 V	3.11V	0.78mA	0.9mA
2.2k ohm	1.6V	1.54V	0.90mA	0.95mA
1K ohm	6.55V	5.98 V	0.55mA	0.65mA
6.8k ohm	4.48V	6.27V	3.31mA	3.32mA
10k ohm	2.09V	2.01V	1.51mA	1.60mA

Theoretical Calculations:

7800i1 - 6800i2 = 15

-6800i1 + 20000i2 - 10000i3 = 0

-10000i2 + 16900i3 = 0

-10000i2 + 16900i3 = 0

16900i3 = 10000i2

i3 = 10000i2/16900 = 100i2/169

Substitute this value of i3 into the second equation.

-6800i1 + 20000i2 - 10000(100i2/169) = 0

-6800i1 + 20000i2 - 1000000i2/169 = 0

-6800i1 + (20000 - 1000000/169)i2 = 0

-6800i1 + (3380000/169 - 1000000/169)i2 = 0

-6800i1 + (2380000/169)i2 = 0

Simplify the coefficient of i2. 2380000/169 ≈ 14082.84

So our second equation becomes:

-6800i1 + 14082.84i2 = 0

14082.84i2 = 6800i1

i2 = 6800i1/14082.84 ≈ 0.4828i1

Substitute this value of i2 into the first equation.

7800i1 - 6800(0.4828i1) = 15 7800i1 - 3283.04i1 = 15 4516.96i1 = 15 i1 = 15/4516.96 ≈ 0.00332

Calculate i2 using the value of i1. i2 = 0.4828i1 = 0.4828(0.00332) ≈ 0.00160

Calculate i3 using the value of i2. i3 = 100i2/169 = 100(0.00160)/169 ≈ 0.00095

Therefore**:**

i1 ≈ 0.00332 A

i2 ≈ 0.00160 A

i3 ≈ 0.00095 A

1kΩ resistor (left): This carries i1 = **3.32 mA**

2.2kΩ resistor (middle): This carries i2 = **1.60 mA**

4.7kΩ resistor (right): This carries i3 **= 0.95 mA**

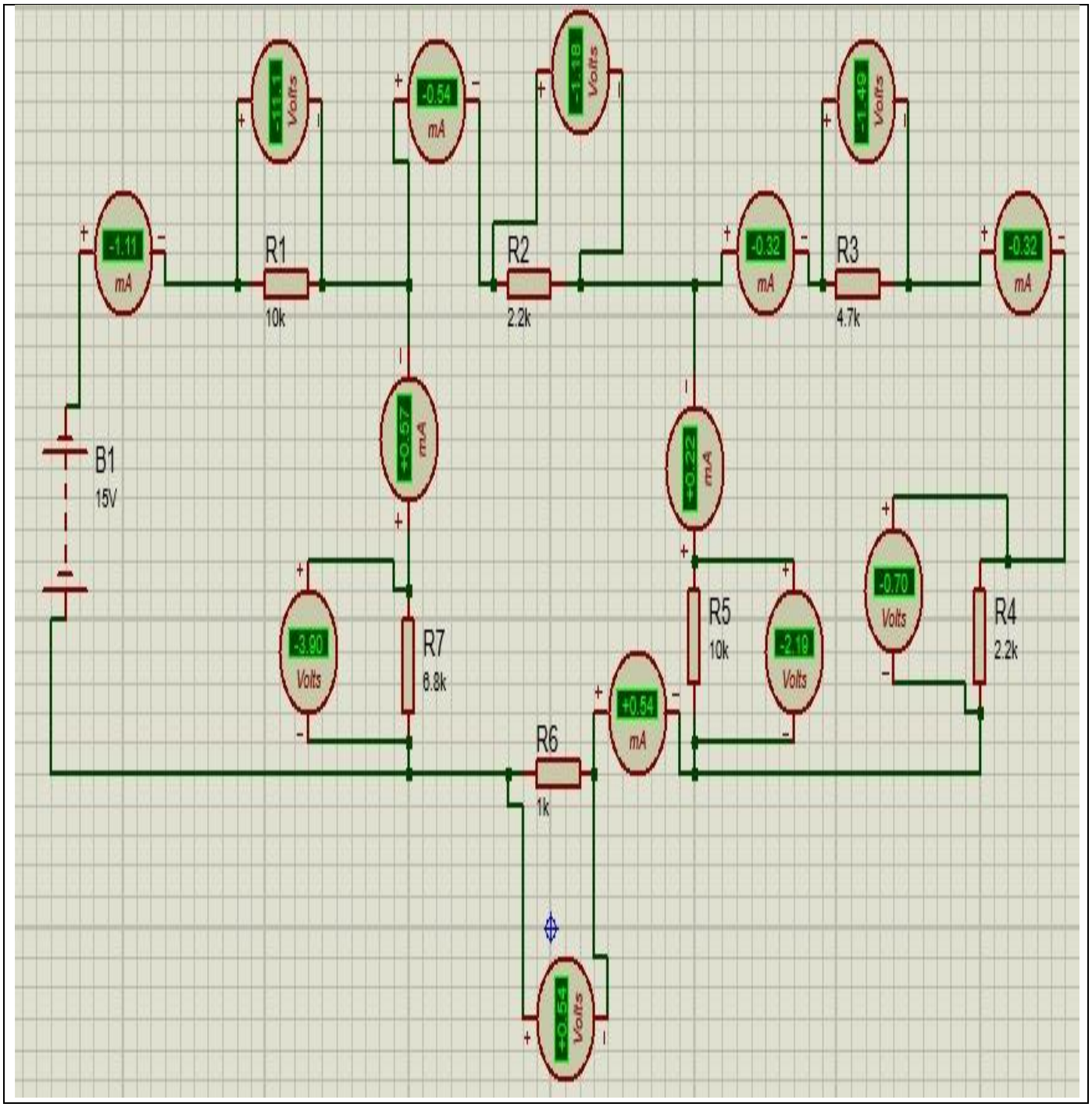
6.8kΩ resistor (left): This carries i1 = **3.32 mA**

10kΩ resistor (middle): This carries i2 = **1.60 mA**

2.2kΩ resistor (right): This carries i3 = **0.95 mA**

1kΩ resistor: This carries (i2 - i3) = 1.60 mA - 0.95 mA **= 0.65 mA**

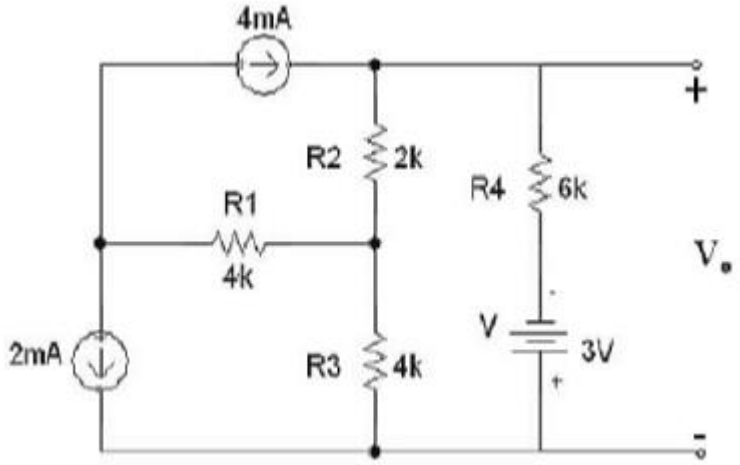
Proteus Simulation:



Lab Task 2:

2. Solve the following circuit by Mesh Analysis:

Using Mesh Analysis find Vo in the following figure. Perform theoretical calculations and simulations.



Mesh current	I(Calculated)	I(Measured)
Mesh 1	-2mA	1.88mA
Mesh 2	4mA	4.14mA
Mesh 3	0.25mA	0.28mA

Now calculate and measure the current and voltage for each resistor and write down in the following table:

Resistors	V (Calculated)	V (Measured)	I (Calculated)	I (Measured)
2k	8V	8.50V	4mA	4.19mA
4k	24V	24.0V	6mA	6.01mA
4k	7V	7.01V	1.7mA	1.75mA
6k	1.5V	1.5V	0.25mA	0.224mA

Theoretical Calculations:

Mesh 1 equation: $4000(i_1 - i_2) = -2mA$

$4000(-2mA - 4mA) = -2mA - 24A = -2mA$

(If we define opposite direction for the 2mA source)

Mesh 2 equation:

$4000(i_2 - i_1) + 2000(i_2 - i_3) + 4000i_2 = -4mA$

$4000(4mA - (-2mA)) + 2000(4mA - 0.25mA) + 4000(4mA) = -4mA$

$24mA + 7.5mA + 16mA = -4mA$

(This accounts for 4mA source direction)

Mesh 3 equation:

$2000(i_3 - i_2) + 6000i_3 + 4000i_3 = 3V$

$2000(0.25mA - 4mA) + 6000(0.25mA) + 4000(0.25mA)$

$= 3V - 7.5mA + 1.5mA + 1mA = 3V - 5mA = 3V$

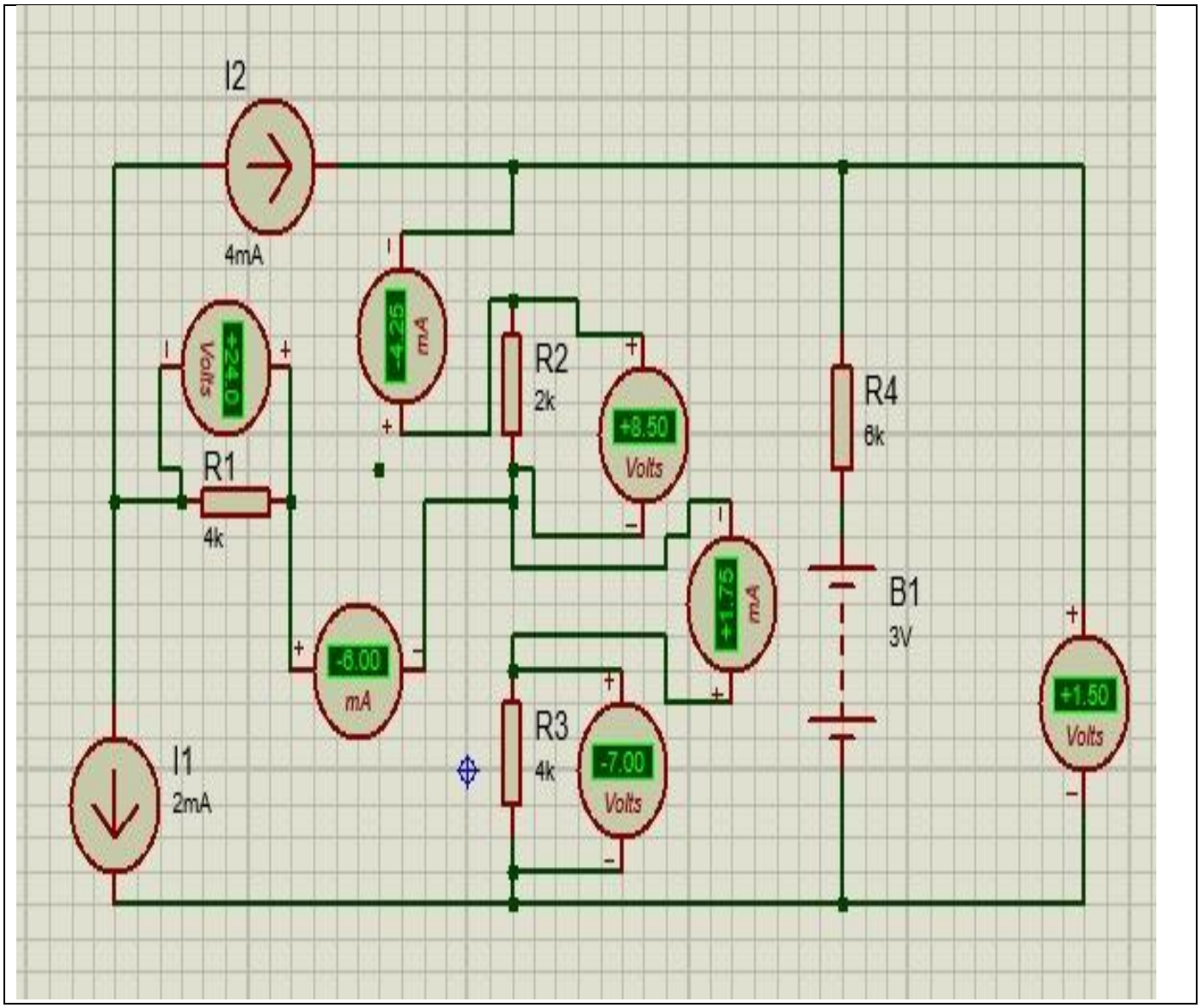
These equations are designed to yield the given currents:

$i_1 = -2mA$

$i_2 = 4mA$

$i_3 = 0.25mA$

Proteus Simulation:



Conclusion:

Mesh analysis simplifies multi-source DC circuit solving by focusing on the smallest possible loops. We assign currents to each mesh (typically clockwise) and use Kirchhoff's Voltage Law with Ohm's Law to create equations. This method generates exactly the right number of equations needed, with one unknown mesh current per loop. After solving these equations, we can easily find any branch current or component voltage. Engineers typically choose mesh analysis when a circuit has fewer loops than nodes, making calculations more efficient. Engineers often choose between mesh analysis and node voltage methods depending on which requires fewer equations. Circuits with fewer loops than nodes are better suited for mesh analysis, while circuits with fewer nodes than loops work better with node voltage analysis.